

Population vs. Sample: The Whole Picture vs. The Smart Slice

Population: The Entire Universe

Includes every member of a group.

It represents the complete set of data you want to study.



**Measured by
"Parameters" (e.g., μ)**

These values, like the true mean, describe the whole population.



**Used when the group
is small and accessible.**

For example, analyzing every purchase at a local bakery.

Sample: The Manageable Snapshot

A subset selected from the population.

It is used when studying the entire population is impractical.



**Measured by
"Statistics" (e.g., $\bar{x}$)**

These values are used to estimate the population's parameters.

**Essential when the group
is large and dispersed.**

For example, surveying 500 teens to understand city-wide trends.



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SAMPLES AND POPULATIONS

These are important words used in statistics:

Population: The whole group of objects or people from whom we are collecting data.

Sample: A group chosen to take part in a survey or to be measured or tested.

Random sample: A sample selected so that any person or object has as much chance as any other of being selected.

Inference: A conclusion you make based on your survey or investigation.

For example, suppose we conduct a survey on how much chocolate students at your school eat. The *population* is the students at your school.

A *sample* is chosen by selecting 10 students at random from the school roll. An *inference* might be that most students eat chocolate at least once a week.



- 3 From a colony of 10 000 ants, 300 are collected and examined for red eye colour. 36 were found to have red eyes.
- a How many ants form the population?
 - b How large was the sample?
 - c What percentage of the sample had red eyes?
 - d Estimate the total number of red-eyed ants.
- 4 50 people were randomly selected from the 750 who attended the opening night of a new play. Of the 50 people, 33 said that they liked the play.
- a What was the population size of people attending the play?
 - b How large was the sample?
 - c What percentage of the sample did *not* like the play?
 - d Estimate the total number of people who did not like the play.

